

Family name: _____ Given name: _____

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1. **Estimate the cost** of the query below (including the writing of the final output, but not any intermediate results) under the following assumptions:¹

- Fact table schema: Sales(TimeID, CustomerID, LocationID, income, expenses)
- Execution strategy: Pipelining
- Memory available: Seven pages (two devoted exclusively to each join operator and one for the final output)
- Join algorithm available: Block Nested Loops
- No indexes, no caching, and no parallelism available
- All attributes are statistically independent and values uniformly distributed
- Any CPU or memory cost can be ignored, only disk access is relevant
- $|Sales| = 10^7$; $B_{Sales} = 10^5$; $R_{Sales} = 100$
- $|Customer| = 10^4$; $B_{Customer} = 200$; $R_{Customer} = 50$; $Dist(Gender) = 2$
- $|Time| = 10^3$; $B_{Time} = 10$; $R_{Time} = 100$; $Dist(Year) = 4$
- $|Location| = 10^3$; $B_{Location} = 10$; $R_{Location} = 100$; $Dist(City) = 10$
- $Cost_{I/O} = 1$

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SELECT s.TimeID, s.CustomerID, s.LocationID, s.income, s.expenses
FROM Sales s, Time t, Customer c, Location l
WHERE s.TimeID = t.ID AND s.CustomerID = c.ID AND s.LocationID = l.ID
AND t.Year = 2023 AND l.City = "Barcelona" AND c.Gender = "Female";
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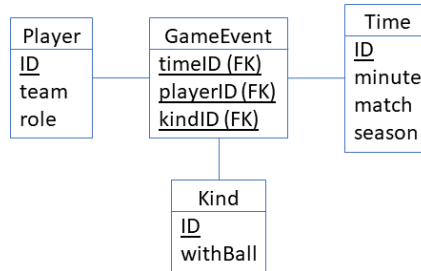
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¹Being $Dist(attribute)$ the number of different values of the attribute.

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2. The multidimensional schema below corresponds to events in football matches with and without ball (e.g., pass, tackling) involving players with any role in the team (e.g., goal-keeper, defender). Indicate the changes you would make in the schema to **record the history** of all the teams of a player given that there is not any maximum limit.



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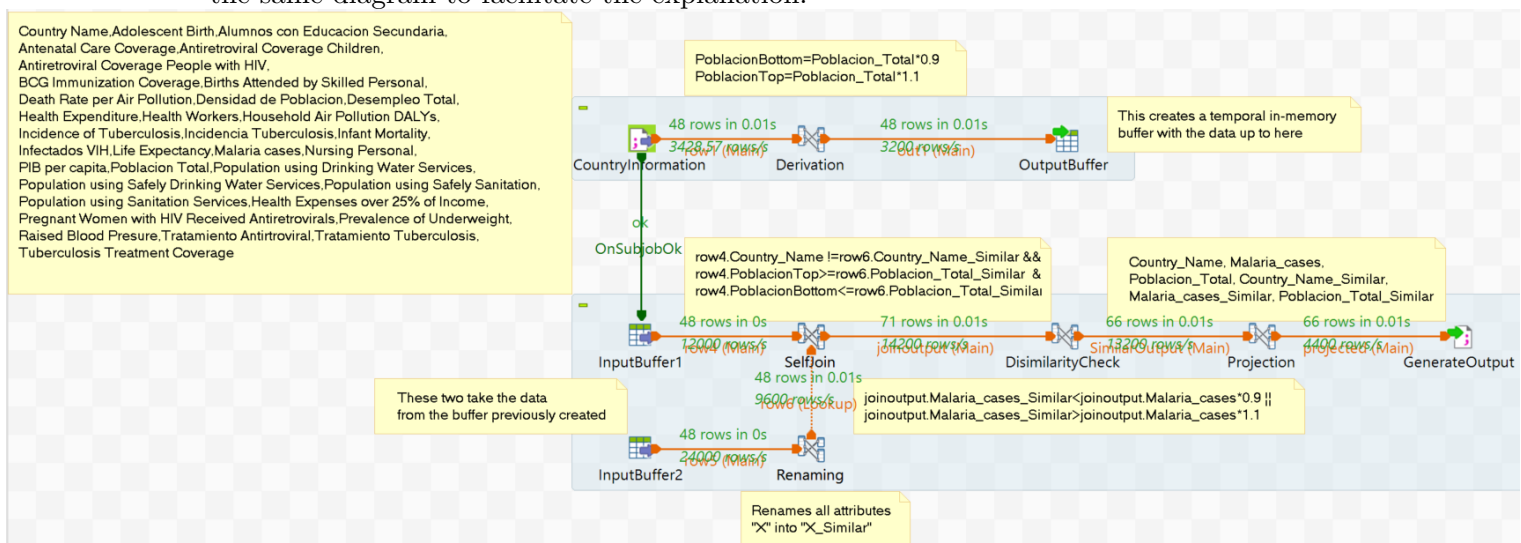
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3. Given the ETL below,^{2,3} where the stickers indicate the attributes and conditions of the nearest node, briefly explain how you would **optimize** it or justify why you think it is already optimal. Explicit any assumption you need to make and annotate any change in the same diagram to facilitate the explanation.



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²The buffer is required in Talend just to have the same data twice in the same operator (i.e., SelfJoin).

³"OnSubjobOK" arrow is just a control flow mechanisms to trigger the second subjob when the first finishes.

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4. Fill the necessary⁴ rows in the follow-up table with the content of sets I and O at the end of the corresponding iteration while applying the R-Swoosh algorithm to the dataset below, using the following:

- Treat all datasets in the algorithm (including the input) as ordered lists
 - Pick/Explore always the rows in the corresponding order
 - Insert always at the end of the list
- Use Hamming distance⁵ over the attributes with threshold 1 as similarity function (i.e., we consider two rows are the same if they have at most one attribute that is different)
- Use “pick the value of the row appearing first in the original dataset” as merge function

Original dataset:

A	B	C
a	1	x
b	1	x
a	2	x
a	1	y

Follow-up table:

Iteration	I	O
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

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5. Consider a hospital with a relation *Patient*(ID, *age*). If we updated yesterday the age of all the patients of the hospital and we assume that the ID also computes for the quality of the table, but never changes, which is the quality of the table from the **timeliness** point of view? There is no need to compute the numeric results, just clearly indicate the calculations and justify them.

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⁴Do not preclude anything because of the number of available rows in the table template.

⁵Hamming($[x_1, \dots, x_n], [y_1, \dots, y_n]$) = $\sum_{i=1}^n val(x_i, y_i)$, with $val(a_1, a_2) = \{ \text{if } (a_1 == a_2) : 0 \text{ else: } 1 \}$